Customer, Product, and Profitability Analysis

APL Logistics (KWE Group)

1. Import Libraries

In [380…

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Load Dataset

In [383…

```
df = pd.read_csv("APL_Logistics.csv", encoding='latin1')
df.head()
```

Out[383…

	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Catego
0	DEBIT	6	4	159.69	472.45	Late delivery	1	
1	DEBIT	4	4	48.71	167.96	Shipping on time	0	
2	DEBIT	4	4	87.36	181.99	Shipping on time	0	
3	DEBIT	6	4	-41.89	175.99	Late delivery	1	
4	DEBIT	6	4	10.00	40.00	Late delivery	1	

5 rows × 40 columns

3. Data Understanding

In [385…

```
df.shape
df.columns
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180519 entries, 0 to 180518
Data columns (total 40 columns):
 #   Column                         Non-Null Count   Dtype
---  ------                         --------------   -----
 0   Type                           180519 non-null  object
 1   Days for shipping (real)       180519 non-null  int64
 2   Days for shipment (scheduled)  180519 non-null  int64
 3   Benefit per order              180519 non-null  float64
 4   Sales per customer             180519 non-null  float64
 5   Delivery Status                180519 non-null  object
 6   Late_delivery_risk             180519 non-null  int64
 7   Category Id                    180519 non-null  int64
 8   Category Name                  180519 non-null  object
 9   Customer City                  180519 non-null  object
 10  Customer Country               180519 non-null  object
 11  Customer Fname                 180519 non-null  object
 12  Customer Id                    180519 non-null  int64
 13  Customer Lname                 180511 non-null  object
 14  Customer Segment               180519 non-null  object
 15  Customer State                 180519 non-null  object
 16  Customer Street                180519 non-null  object
 17  Customer Zipcode               180516 non-null  float64
 18  Department Id                  180519 non-null  int64
 19  Department Name                180519 non-null  object
 20  Latitude                       180519 non-null  float64
 21  Longitude                      180519 non-null  float64
 22  Market                         180519 non-null  object
 23  Order City                     180519 non-null  object
 24  Order Country                  180519 non-null  object
 25  Order Customer Id              180519 non-null  int64
 26  Order Item Discount            180519 non-null  float64
 27  Order Item Discount Rate       180519 non-null  float64
 28  Order Item Product Price       180519 non-null  float64
 29  Order Item Profit Ratio        180519 non-null  float64
 30  Order Item Quantity            180519 non-null  int64
 31  Sales                          180519 non-null  float64
 32  Order Item Total               180519 non-null  float64
 33  Order Profit Per Order         180519 non-null  float64
 34  Order Region                   180519 non-null  object
 35  Order State                    180519 non-null  object
 36  Order Status                   180519 non-null  object
 37  Product Name                   180519 non-null  object
 38  Product Price                  180519 non-null  float64
 39  Shipping Mode                  180519 non-null  object
dtypes: float64(13), int64(8), object(19)
memory usage: 55.1+ MB
```

Out[385…

	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Late_delivery_risk
count	180519.000000	180519.000000	180519.000000	180519.000000	180519.000000
mean	3.497654	2.931847	21.974989	183.107607	0.548291
std	1.623722	1.374449	104.433526	120.043668	0.497664
min	0.000000	0.000000	-4274.980000	7.490000	0.000000
25%	2.000000	2.000000	7.000000	104.380000	0.000000
50%	3.000000	4.000000	31.520000	163.990000	1.000000
75%	5.000000	4.000000	64.800000	247.400000	1.000000
max	6.000000	4.000000	911.800000	1939.990000	1.000000

8 rows × 21 columns

Dataset contains ~180k records and 40 columns including sales, profit, customer, and logistics data.

4. Data Cleaning & Financial Validation

In [389…

```
# Missing values
df.isnull().sum()

# Drop duplicates
df = df.drop_duplicates()

# Remove invalid data
df = df[df['Sales'] > 0]
df = df[df['Order Profit Per Order'] != 0]

# Fill missing values
df = df.fillna(0)
```

Removed duplicate records and invalid financial entries to ensure accurate analysis.

5. Create KPI Columns

In [393…

```
df['Profit Margin'] = (df['Order Profit Per Order'] / df['Sales']) * 100
df['Discount Impact'] = df['Order Item Discount'] / df['Sales']
```

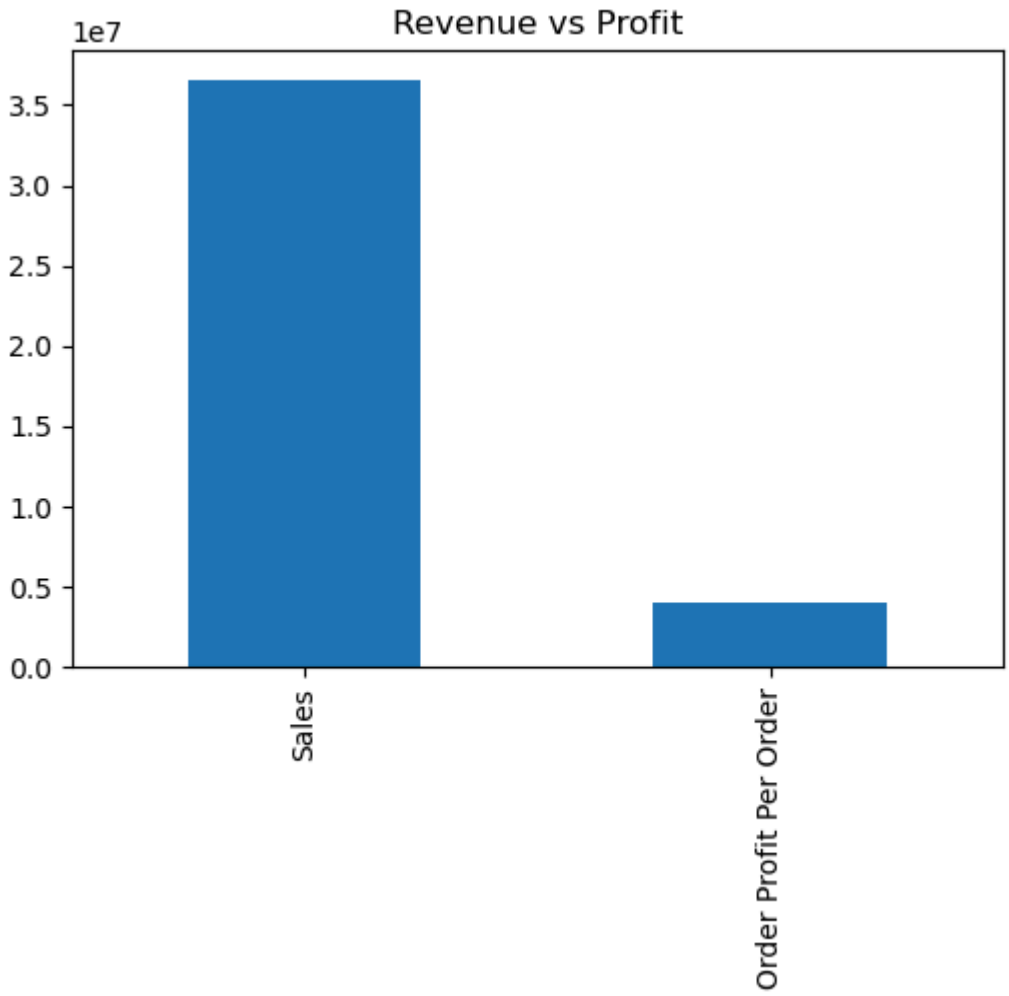
6. Revenue & Profit Overview

In [396…

```
total_sales = df['Sales'].sum()
total_profit = df['Order Profit Per Order'].sum()
profit_margin = (total_profit / total_sales) * 100

total_sales, total_profit, profit_margin
```

Out[396… (36552703.230000004, 3966902.97, 10.852557046298651)

In [399…
```
# 📊 Visualization:
df[['Sales', 'Order Profit Per Order']].sum().plot(kind='bar', figsize=(6,4))
plt.title("Revenue vs Profit")
plt.show()
```



Revenue is high, but profit margin is relatively moderate, indicating inefficiencies.

7. Product & Category Profitability Analysis

In [403…
```
customer_df = df.groupby('Category Name').agg({
    'Sales': 'sum',
    'Order Profit Per Order': 'sum'
}).reset_index()

category_df['Margin %'] = (
    category_df['Order Profit Per Order'] / category_df['Sales']
) * 100

category_df.sort_values(by='Margin %', ascending=False)
# Loss-making categories
loss_categories = category_df[category_df['Margin %'] < 0]
loss_categories
# High sales but low margin
high_sales_low_margin = category_df[
    (category_df['Sales'] > category_df['Sales'].mean()) &
```

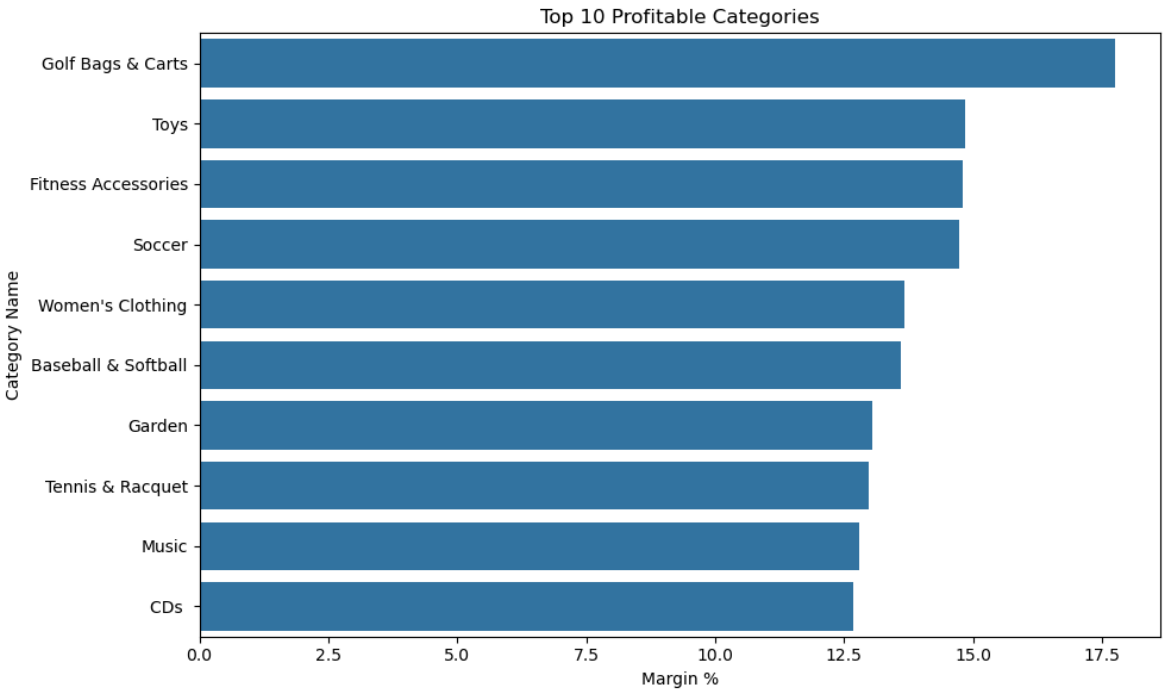
```
    (category_df['Margin %'] < category_df['Margin %'].mean())
]

high_sales_low_margin
```

Out[403…

	Category Name	Sales	Order Profit Per Order	Margin %
9	Camping & Hiking	4091727.20	427455.57	10.446825
10	Cardio Equipment	3669885.66	383011.10	10.436595
38	Shop By Sport	1300744.17	129813.96	9.979976
46	Water Sports	3095645.51	325146.96	10.503365

In [405…

```
# 📊 Visualization:
top_categories = category_df.sort_values(
    by='Margin %', ascending=False
).head(10)

plt.figure(figsize=(10,6))

sns.barplot(
    data=top_categories,
    x='Margin %',
    y='Category Name'
)

plt.title("Top 10 Profitable Categories")
plt.tight_layout()
plt.show()
```



- Top categories show higher profitability, while some categories have lower margins.
- Certain categories generate high revenue but have low profit margins, indicating pricing or cost issues.

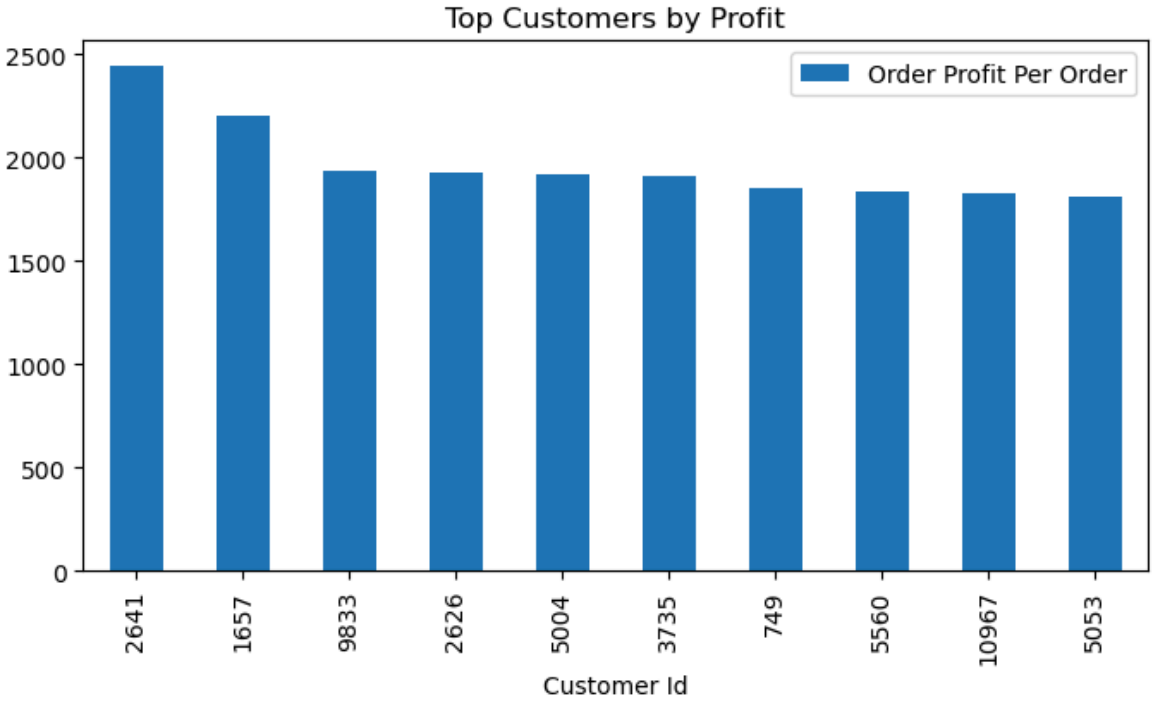
8. Customer Contribution Analysis

In [457…

```
customer_df = df.groupby('Customer Id').agg({
    'Sales': 'sum',
    'Order Profit Per Order': 'sum'
}).reset_index()

customer_df = customer_df.sort_values(by='Order Profit Per Order', ascending=Fal
```

In [459…

```
# 📊 Chart
customer_df.head(10).plot(
    x='Customer Id',
    y='Order Profit Per Order',
    kind='bar',
    figsize=(8,4)
)
plt.title("Top Customers by Profit")
plt.show()
```



9. Customer Segmentation

In [452…

```
customer_df['Segment'] = pd.qcut(
    customer_df['Order Profit Per Order'],
    q=3,
    labels=['Low Value', 'Medium Value', 'High Value']
)

customer_df['Segment'].value_counts()
```

Out[452…

```
Segment
Medium Value    6867
Low Value       6866
High Value      6866
Name: count, dtype: int64
```

- A small group of customers contributes most of the profit
- Some customers generate losses despite high sales

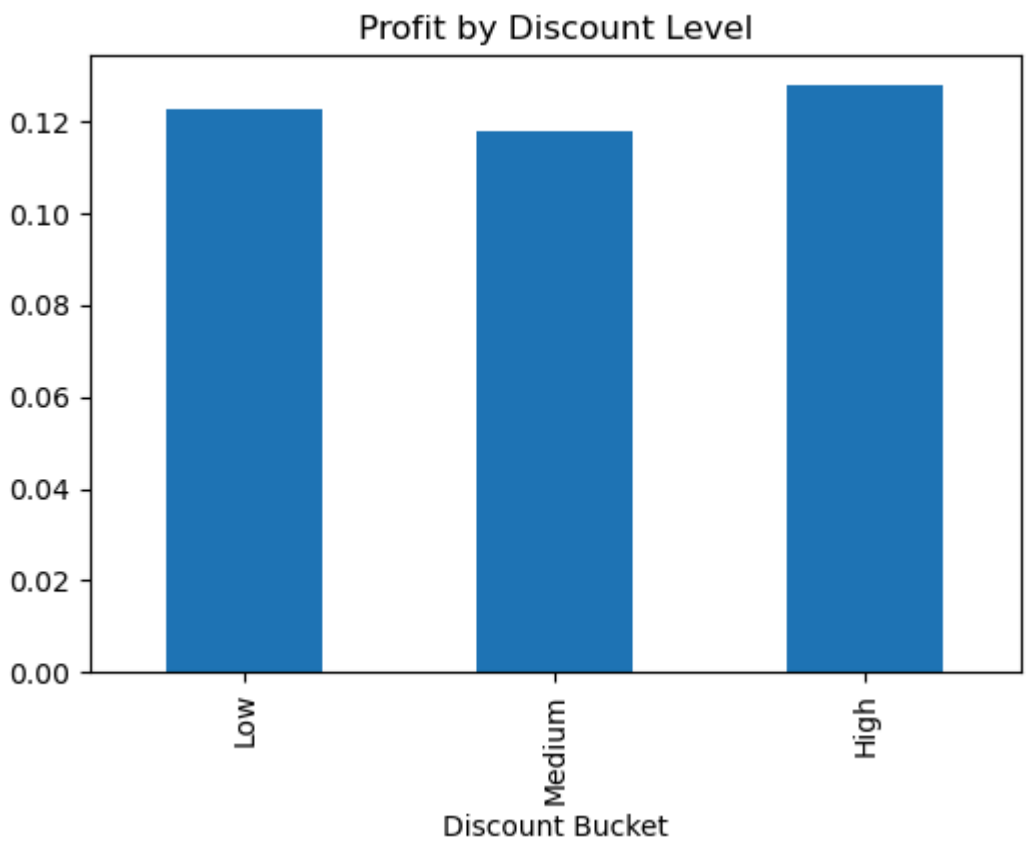
10. Discount Impact Diagnostics

In [418…

```
plt.figure(figsize=(8,5))

sns.regplot(
    x=df['Order Item Discount Rate'],
    y=df['Order Item Profit Ratio'],
    scatter_kws={'alpha':0.3}
)

plt.title("Discount vs Profit (Trend)")
plt.show()
```

In [419…

```
# Discount threshold analysis
df['Discount Bucket'] = pd.cut(
    df['Order Item Discount Rate'],
    bins=[0, 0.1, 0.2, 0.3],
    labels=['Low', 'Medium', 'High']
)

discount_analysis = df.groupby('Discount Bucket', observed=True)['Order Item Pro
discount_analysis
discount_analysis.plot(kind='bar', figsize=(6,4))
plt.title("Profit by Discount Level")
plt.show()
```



In [420…

```
discount_analysis_full = df.groupby('Discount Bucket', observed=True).agg({
    'Order Item Profit Ratio': 'mean',
    'Sales': 'count'
})

discount_analysis_full
```

Out[420…

	Order Item Profit Ratio	Sales
Discount Bucket		
Low	0.122558	89660
Medium	0.118070	69764
High	0.127949	9967

Discount Impact Analysis

- Medium discount levels show the lowest profitability, indicating inefficient pricing strategies.
- High discount transactions appear profitable but occur at a much lower volume.
- Low discount levels contribute the majority of orders with stable profit margins.
- This suggests that excessive discounting is not consistently beneficial and must be applied strategically.

11. Market & Regional Profit Analysis

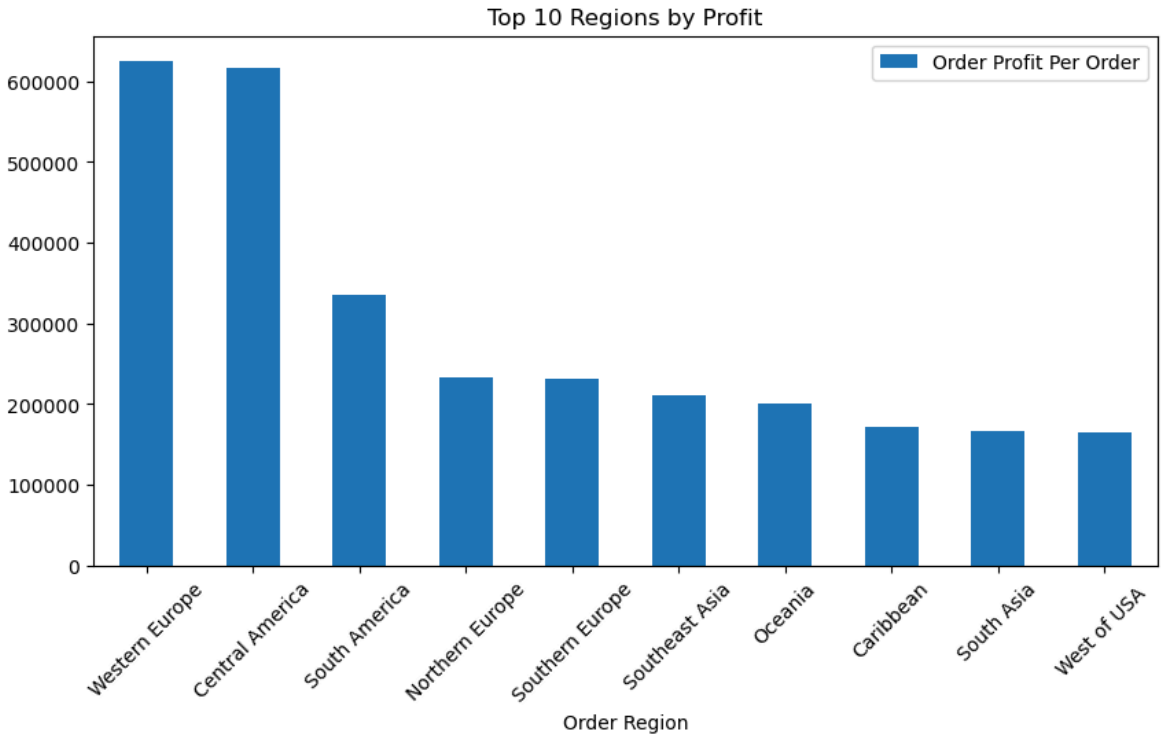
In [423…

```
region_df = df.groupby('Order Region').agg({
    'Sales': 'sum',
```

```
    'Order Profit Per Order': 'sum'
}).reset_index()
```

In [424…

```
# 📊 Visualization:
top_regions = region_df.sort_values(
    by='Order Profit Per Order',
    ascending=False
).head(10)

top_regions.plot(
    x='Order Region',
    y='Order Profit Per Order',
    kind='bar',
    figsize=(10,5)
)

plt.title("Top 10 Regions by Profit")
plt.xticks(rotation=45)
plt.show()
```



- Some regions show strong revenue but weak profitability
- Indicates operational inefficiencies or high costs

12. Key Performance Indicators (KPIs)

In [427…

```
# 12. KPI Summary
# Customer Value Index
customer_value = df.groupby('Customer Id')['Order Profit Per Order'].sum()
customer_value_index = customer_value.mean()

# Category Margin
category_df = df.groupby('Category Name').agg({
    'Sales': 'sum',
    'Order Profit Per Order': 'sum'
```

```
}).reset_index()

category_df['Margin %'] = (
    category_df['Order Profit Per Order'] / category_df['Sales']
) * 100

category_margin = category_df['Margin %'].mean()

# Discount Impact Ratio
discount_impact_ratio = (
    df['Order Item Discount'].sum() / df['Sales'].sum()
) * 100

# KPI Dictionary
kpi = {
    "Total Revenue": df['Sales'].sum(),
    "Total Profit": df['Order Profit Per Order'].sum(),
    "Profit Margin (%)": (df['Order Profit Per Order'].sum() / df['Sales'].sum()
    "Customer Value Index": customer_value_index,
    "Category Margin (%)": category_margin,
    "Discount Impact Ratio (%)": discount_impact_ratio
}

kpi
```

Out[427…

```
{'Total Revenue': 36552703.230000004,
 'Total Profit': 3966902.97,
 'Profit Margin (%)': 10.852557046298651,
 'Customer Value Index': 192.57745375989126,
 'Category Margin (%)': 10.832291048558007,
 'Discount Impact Ratio (%)': 10.144102466700106}
```

KPI Insights

- The company generated total revenue of 36.55M with a profit of 3.96M.
- Profit margin is approximately 10.85%, indicating moderate efficiency.
- Customer Value Index (~192) shows strong average profit per customer.
- Category Margin (~10.83%) suggests consistent profitability across products.
- Discount Impact (~10.14%) indicates discounts significantly affect margins.

In [429…

```
pd.DataFrame(list(kpi.items()), columns=['KPI', 'Value'])
```

Out[429…

	KPI	Value
0	Total Revenue	3.655270e+07
1	Total Profit	3.966903e+06
2	Profit Margin (%)	1.085256e+01
3	Customer Value Index	1.925775e+02
4	Category Margin (%)	1.083229e+01
5	Discount Impact Ratio (%)	1.014410e+01

13. Final Insights & Recommendations

Key Insights

- High revenue does not guarantee high profit
- Discounts significantly reduce profitability
- Few customers contribute majority of profit
- Some categories have lower margins
- Regional performance varies significantly

Recommendations

- Optimize discount strategies
- Focus on high-value customers
- Improve pricing for low-margin products
- Strengthen operations in weak regions

In [ ]:
```
df.to_csv("APL_Logistics_Cleaned.csv", index=False)
```

In [ ]:

In [ ]: